

INMO(19th) – 2004

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Consider a convex quadrilateral ABCD in which K, L, M, N are the midpoints of the sides AB, BC, CD, DA respectively. Suppose:
 - (a) BD bisects KM at Q.
 - (b) QA = QB = QC = QD
 - (c) LK/LM = CD/CB.
 Prove that ABCD is a square.
 2. Suppose p is a prime greater than 3. Find all pairs of integers (a, b) satisfying the equation $a^2 + 3ab + 2p(a + b) + p^2 = 0$.
 3. If α is a root of the equation $x^5 - x^3 + x - 2 = 0$, prove that $[\alpha^6] = 3$ where $[x]$ denotes the greatest integer function.
 4. Let R denotes the circum radius of a triangle ABC; a, b, c its sides BC, CA, AB; and r_a, r_b, r_c its exradii opposite A, B, C. If $2R \leq r_a$, prove that
 - (a) $a > b$ and $a > c$.
 - (b) $2R > r_b$ and $2R > r_c$.
 5. Let S denote the set of 6-tuples (a, b, c, d, e, f) of positive integers such that $a^2 + b^2 + c^2 + d^2 + e^2 = f^2$. Consider the set $T = \{abcdef : (a, b, c, d, e, f) \in S\}$. Find the greatest common divisor of all the members of T .
 6. Prove that the number of 5-tuples of positive integers (a, b, c, d, e) satisfying the equation $abcde = 5(bcde + acde + abde + abce + abcd)$ is an odd integer.

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